Project Title	Customer Churn and Feedback Sentiment Prediction Project
Skills take away From This Project	Data cleaning, extraction, feature selection, ML, DL and NLP for sentiment analysis. Streamlit for dashboard preparation
Domain	Customer rentention, Customer Analytics

Problem Statement:

Customer Churn and Feedback Sentiment Prediction using Machine Learning and Deep Learning. The goal of this project is to analyze a customer datasets containing both structured (tabular) information and unstructured (textual feedback) data to predict customer churn and understand sentiment patterns in customer feedback. Learners will build an end-to-end pipeline from data preprocessing to model evaluation and deployment-ready solution.

Business Use Cases:

This project's solution can be applied in several real-world business scenarios:
- Telecom / SaaS / Banking / E-commerce: Predict which customers are likely to churn based on their profile, usage patterns, and feedback text.
- Customer Experience Teams: Identify negative sentiment in feedback to proactively resolve issues before churn.
- Marketing and Retention Teams: Target high-risk customers with retention campaigns based on churn prediction scores.
- Product Management: Derive actionable insights from text feedback to improve products and services.

Approach:

1. Data Understanding and Cleaning
2. Feature Engineering
2. Visualization & Insights
3. Model Development (ML & DL)

4. Model Validation & Evaluation
5. NLP Processing by using LSTM model

Results:

By the end of the project, learners should be able to:
- Successfully predict customer churn using both ML and DL techniques.
- Analyze customer feedback using NLP to identify major pain points.
- Deliver insights that could help reduce churn and improve customer satisfaction.
- Optionally develop a small dashboard to visualize predictions.

Project Evaluation metrics:

Model Accuracy
- Precision, Recall, F1-Score
- ROC-AUC
- NLP Model Performance
- Code Quality
- Documentation & Insights

GUVI | HCL
Skill Up. Level Up

Technical Tags:

DataScience, MachineLearning, DeepLearning, NaturalLanguageProcessing, CustomerAnalytics, FeatureEngineering, SentimentAnalysis.

Data Set:

Dataset Name: Customer Churn Prediction

https://drive.google.com/drive/folders/1HldyzpKLA_R2zUMgWf6t93A0PbwSbBYb?usp=sharing

Merge all the datasets and prepare a common one and proceed with the solution. You can consider other related datasets as well.

Project Deliverables:

- Source Code (preprocessing, ML, DL, NLP scripts)
- Model Artifacts (pickle/h5 files)
- Documentation (README, insights report)
- Optional dashboard (streamlit) /API(Flask or FastAPI) for predictions
- Deploy the model through API either in Lambda or EC2

Project Guidelines:

Use version control (Git/Github)
- Follow PEP8 coding standards
- Use modular code (functions/classes)
- Validate models with cross-validation
- Ensure reproduceability with fixed seeds and requirements.txt

Timeline:

2 Weeks

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (4:00PM to 5:00PM)

Booking link :https://forms.gle/XC553oSbMJ2Gcfug9

For DE/BADM project/class topic doubt slot clarification session:

Booking link : https://forms.gle/NtkQ4UV9cBV7Ac3C8

Session timing:

For DE: 04:00 pm to 5:00 pm every saturday
For BADM 05:00 to 07:00 pm every saturday

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For BADM and DE
Monday-Saturday (11:30AM to 1:00PM)

For DS and AIML
Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

Created By:	Verified By:	Approved By:

GUVI | HCL
Skill Up. Level Up